

Review of Brain Tumor MRI Image Segmentation Methods for BraTS Challenge Dataset

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Abstract—Brain Tumor MRI segmentation is a crucial task in biomedical imaging. Early discovery of brain cancer can help with improving the quality of life and survivability post-treatment. In the case of children affected with brain tumors, early detection can determine what therapy would be required and early treatment in most cases will increase the longevity of their life. Brain Tumor segmentation has been done manually by an expert operator who is clinically trained. But this is very time-consuming and also the rating by these trained operators has intra-operator variability in most cases. Also, the different machines from different places have few variations in imaging which brings the rating variations. A need for Fully automated segmentation models is required for overcoming these issues. In recent years we have seen a trend of deep learning models being in heavy use in medical imaging tasks. These deep learning models have exhibited state-of-the-art performance by self-learning features. In this paper we focus on different machine learning, deep learning models being used on the Brain Tumor Segmentation (BraTS) challenge datasets. This paper tries to give the overall work-flow required for Brain Tumor MRI segmentation and gives a comparison on different models centered around deep learning as well as machine learning models. Lastly, an evaluation of the present state is shown and future improvements to standardize MRI-based brain tumor segmentation techniques.

Index Terms—Brain Tumor Segmentation, BraTS, U-Net, CNN, ResNet, CycleGan

I. INTRODUCTION

A brain tumor is a collection of an unusual mass of cells in your brain. Your skull, which includes your brain, is very rigid. Because of this rigidity, any form of growth in the brain can cause abnormalities in the proper functioning of the brain. The way a brain tumor can affect the body is interesting. A brain tumor sometimes may not disturb the body's functions. It is very uncommon for a brain tumor that is malignant to grow way out of proportion and protrude outside of the brain. Brain tumors affect both the brain or the spinal cord wherein they compress parts of these and compromise the related brain functions. The brain in conjunction with the spinal cord controls and is accountable for the coherence of our body coordination. The primary area that affects our movement, coordination, and regulation may not be the area where the brain tumor may have originated primarily it may be a collateral effect. Based on the location of the tumor, they are categorized into Benign, Malignant, Primary, and Metastatic. Given the different types of brain tumors, the one that we focus

on here is known as the Gliomas which are a type of brain tumor that starts from the glial cells [1]. Brain tumor MRI segmentation has been partitioned into three groups depending on the degree of engagement of the human in segmenting the scans. It can be categorized into Manual or fully automatic segmentation [2] and also there are semi-automatic methods [3]. Manual segmentation is an incredibly time-intensive task and is very much subject to changes depending on the person who is segmenting it manually. There are times when the different rater doesn't agree with each other regarding the segmentation result. A major challenge can be the automatic segmentation of gliomas. A 3D image of the Tumor-bearing brain's image is at hand whose volume and placement can differ significantly from patient to patient. Especially a limitation that comes into play with traditional edge-based methods is the large indistinct and irregular tumor boundaries with discontinuities. Additionally, brain tumor MRI data acquired is intrinsically complex when obtained clinically or through synthetic methods. In Indian on average, 5 to 10 per 100,000 population is the fraction of central nervous system (CNS) tumors with an increasing change [4]. Out of these, 40 % of them propagate to the brain. Every year 40,000-50,000 persons are identified with a brain tumor in India [21]. Twenty Percent of these are children with brain tumor. An early accessible prognosis is necessary for proper treatment in such cases.

II. MOTIVATION FOR RESEARCH

Manual segmentation of Brain MRI is very laborious and has sometimes varying results which hinge on the operator who annotates the MRI. And also in most of the time, Multiple MRI modalities are not used because of the limitation that comes with doing it physically. So This paper's main purpose is to review the various methods which have made an impact in the research area of brain tumor segmentation techniques that are semi or fully automatic. This review paper intends at making this paper in such a way so that most of the components in this area can be explained briefly and state-of-the-art methods are also presented. The dataset here has been focused mainly on the BraTS [5], [6], [7] challenge dataset which has been happening since 2012 and all the paper that has been reviewed has either primarily used BraTS or have their model tested on BraTS dataset.

TABLE I: Summary of the Brain Tumor Segmentation (BraTS) challenge dataset from 2012 To 2020

Year	Dataset			Annotation Method
	Training Set	Validation Set	Test Set	
BraTS 12	30 clinical data + 50 Synthetic Data	N/A	15 Clinical data + 15 Synthetic Data	Manual annotation by expert raters(Two tumor class: Edema and core)
BraTS 13	Clinical Data from BraTS 12 Training set	N/A	15 clinical data from BraTS 12 + 10 New test data	Manual annotation by expert raters(Four tumor classes)
BraTS 14	200 datasets from BraTS 12, 13 and TCIA	N/A	38 Unseen data from BraTS 12, 13 test set and TCIA	Using results of high-ranked methods from BraTS 12 and BraTS 13 and getting approved by expert raters.
BraTS 15	Same as BraTS 14 training set	N/A	53 Unseen data from BraTS 12, 13 test set and TCIA	Same Method as BraTS 14
BraTS 16	Same as BraTS 15 training set	N/A	191 Unseen data from BraTS 12, 13 test set and TCIA	Same Method as BraTS 15
BraTS 17	285 datasets from BraTS12 and BraTS13 training + Large dataset from 19 institutions	46 unseen datasets from different institutes	146 unseen data from BraTS 13 test set + Different institute data	BraTS 12 and 13 data manually annotated + 19 institute MRI scans
BraTS 18	Same as BraTS 17	66 unseen data from different institutes	191 unseen data from BraTS 13 test set + Different institutes	Same as BraTS 17
BraTS 19	335 including from BraTS 17	125 unseen data from different institutes	From Previous test sets	Same As BraTS 18
BraTS 20	269 including from BraTS 19	125 unseen data from different institutes	From Previous test sets	Same As BraTS 19

III. BRATS DATASET SPECIFICATION

The inception of the BraTS image segmentation challenge started in 2012 and has been conducted yearly as a crucial component of the MICCAI [5], [6], [7] conference subsequently. Participants sign up for the contest online to gain access to the dataset. The participants make a model implementing the brain tumor MRI segmentation and then they test their model against a test dataset and submit a paper explaining the preliminary results. The organizers give two distinct datasets every year: one would be the training dataset with an annotation which is used for implementing a model and a test data is used for evaluating the segmentation model. Subsequently from BraTS 2017, the challenge datasets consisted of a validation dataset. Contestants do the segmentation model training on the dataset and finally, upload the findings for evaluation to the organizer's server. In 2014, the second job of disease advancement assessment has been added which advanced the BraTS challenge from being a

brain tumor segmentation challenge to include a task that underscores a clinical significance in the challenge. Table I shows the summary of the BraTS dataset through 2012-2020 and show the bifurcation of dataset into train, test, validation sets and also how they were obtained.

IV. REVIEW OF BRAIN MRI SEGMENTATION METHODS

A. Related Surveys

Several related review papers are published on the brain Tumor Segmentation method. For Example, Ghaffari et al. [8] presented a detailed review of methods on the BraTS dataset but on Multiple Sclerosis. And Some other survey papers worth mentioning are [9], [10], [11], [12]. This research area of Brain MRI segmentation is still in its early stages compared to what sort of segmentation is being achieved for natural images.

B. Image Pre-Processing methods

Standard preprocessing stages for brain MRI comprises of the subsequent key steps.

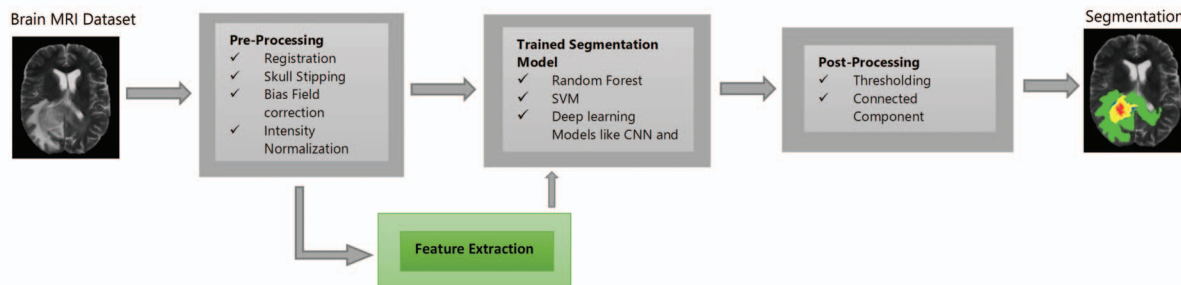


Fig. 1: General Flow of Brain MRI segmentation methods.

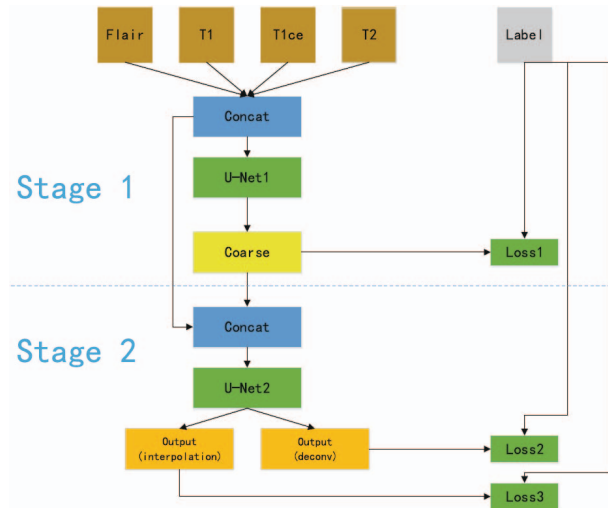


Fig. 5: Overview of Two Stage Cascaded Network [16]

datasets and also uses less training data. The authors have chosen a way to were given and a set of labelled images (L) and a set of Brain tumor MRI scans (T) these unpaired MR scans and labelled data are arbitrarily chosen from BraTS 2015 dataset.

In cycle-1 the network G generates the segmented tumor (l_i^1) from the image (t_i) and then the network F generates an image back (t_i^1) synthetically which is similar to t_i^1 . Similarly in cycle-2 uses the network F to generate the MRI scan (t_i^2) from the image and then use G to generate the labelled image back which is similar to l_i . D_T discriminates between (t_i) and (t_i^2) which was generated above in F . Finally D_L aims to distinguish (l_i^1) and (l_i).

Since the Inception of U-Net, it has been extensively used in Biomedical image segmentation tasks. In this paper, Jiang *et al.* [16] propose a U-Net to segment the substructures of brain tumors from coarser to a finer mask and this is in a two-stage cascaded form. An end-to-end network is trained on the BraTS 2019 training dataset. In this method first, as shown in Fig. 5. The first stage U-Net takes in the MRI images and outputs a very coarse segmentation map. More accurate segmentation of the inputs images is obtained by combining the raw images

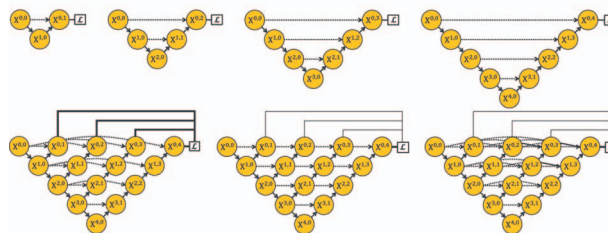


Fig. 6: Evolution from U-Net to U-Net++ [17].

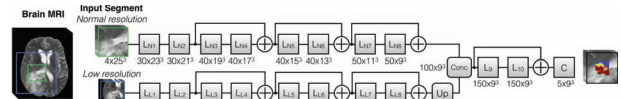


Fig. 7: Deep Medic Architecture [18].

along with the segmentation map and further processing them onto the second stage. The authors use a variation of U-Net which has a Two-Stage Cascaded U-Net. The method used in this paper got the first position in BraTS to challenge 2019. In their method, they use a two-stage cascaded U-Net which is a modification of the U-Net architecture. Further, in the second stage increase the width of the network which enhances the prediction map by stacking an initial prediction map with the original input to make use of auto-context.

Zhou *et al.* [17] proposes U Net++, Semantic and instance segmentation neural architecture where an ensemble of U-Nets of varying depths is used to boost the unknown network depth, here co-learning is done by partly sharing encoder concurrently using deep supervision.

The skip connections in the network are redesigned to accumulate features of contrasting semantic scales at the decoder sub-networks. A flexible feature fusion scheme is developed by formulating a pruning system to accelerate the inference rate of the U-Net++ architecture.

The architectural modifications introduced in U-Net++ facilitate the following enhancements. One reason is that U-Nets of differing depths are less susceptible to the selection of the network depth. These U-Nets use an interlinked decoder with a partially shared encoder. All U-Nets simultaneously get trained by using the method of deep supervision for training U-Net++. Such a design change helps in effective pruning during the inference time while also improving the complete segmentation performance. Most importantly as same scale feature maps from the encoder and decoder are merged so it does not unreasonably restrict skip connections from bottlenecking U-Net++. The Architecture is shown in Fig. 6.

Deep-medic was one of the top-performing algorithms which were proposed by Kamnitsas *et al.* [18]. Deep-medic architecture is a 11-layer 3D CNN model which consists of residual connections as shown in Fig 7. This architecture

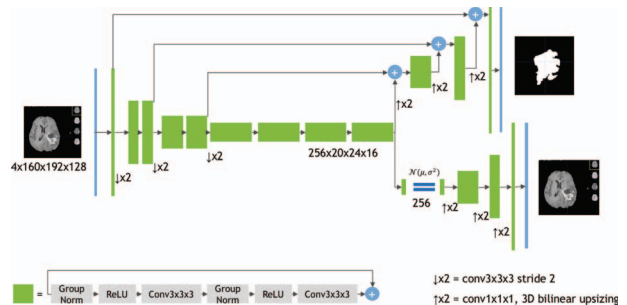


Fig. 8: Asymmetric encoder-decoder architecture [19].

TABLE II: Comparison of the reviewed brain tumor segmentation methods

Reference	Dataset	Method Used	Dice Similarity Coefficient (%)		
			Complete	Core	Enhancing
Urban et al. [13]	BraTS 2013	3D Volumetric CNN	87.00 %	77.00 %	33.00 %
Tustison et al. [14]	BraTS 2013	Concatenated Random Forest	88.00 %	76.00 %	55.00 %
Wang et al. [15]	BraTS 2017	Cascaded Anisotropic CNN	90.21 %	85.83 %	79.72 %
Nema et al. [20]	BraTS 2017	Residual cyclic unpaired encoder-decoder network	94.63 %	85.60 %	93.54 %
Zhou et al. [17]	BraTS 2013	U-Net++: Redesigning Skip Connections to Exploit Multiscale Features	91.21 %	83.00 %	71.00 %
Jiang et al. [16]	BraTS 2019	Two-Stage Cascaded U-Net	90.81 %	86.32 %	80.19 %
Kamnitsas et al. [18]	BraTS 2015	Deep-Medic	89.60 %	76.30 %	72.40 %
Myronenko et al. [19]	BraTS 2018	3D MRI Segmentation Using Autoencoder Regularization	91.05 %	86.60 %	82.30 %

processes the input through two pathways and each of them does it in its specific way. Boosting the receptive field of the final classification while maintaining a low computational cost was a priority. The improved results of the model were due to the residual connections used.

Myronenko et al. [19] proposed a method which is a two-part encoder-decoder based CNN where the first part obtains image feature by using a large asymmetrical encoder and the second part reconstructs the segmentation mask by using a small encoder. The authors tackle the limited training dataset by adding an auto-encoder branch which gives additional guidance and regularization. To cluster or group the attributes of the encoder endpoints the authors follow an approach similar to the variational auto-encoder (VAE). The components of the architecture are shown in above Fig. 8. When the batch size is small Group normalization is used for normalizing instead of Batch-norm because of better performance, here in this model batch size is 1. The decoder and encoder are comparable but only consist of a single block per spatial level.

V. EVALUATION OF MODELS

Different evaluation metrics are used to evaluate how well the model has performed the task of segmentation. These evaluation metrics are also used to find out how good one model is compared to other Brain MRI segmentation methods. The entries in the BraTS challenge are benchmarked and ranked according to two evaluation metrics used by them, one is the Dice Similarity coefficient (DSC) and the other is the Hausdorff Distances

A. Dice similarity coefficient (DSC)

The Dice score is regularly used in research to measure the performance of image segmentation methods. In DSC you mark some ground truth region in your image and then make an automated algorithm to do it.

$$DSC = \frac{2 \times TP}{2TP + FP + FN} \quad (1)$$

Here TP is the true positive which is the pixels in the Ground Truth. False-positive (FP) is the area in which the prediction claims to be correctly classified but it is wrong. False-negative (FN) is the area that is predicted to be not part of the ground truth but actually, it is. So the Dice Score Coefficient gives a value between 0 and 1 and it shows us the 2* Area of overlap divided by the total pixels combined. The Dice is the same as the F1 score.

B. Hausdorff distance

Hausdorff distance is used to find the distance among segmentation boundaries. A set of the shortest least square distance $d(p, t)$ from all points on the predicted label surface ($P \in \partial P$) to the points on the ground truth label ($T \in \partial T$) are created and the maximum value of these are calculated.

$$HD(P, T) = \left\{ \sup_{p \in \partial P} \inf_{t \in \partial T} d(p, t), \sup_{t \in \partial T} \inf_{p \in \partial P} d(t, p) \right\} \quad (2)$$

VI. RESULTS AND DISCUSSION

Table II shows the results of the reviewed models on BraTS dataset. A good cascaded CNN method as proposed by Wang et al. [15] shows promising results by dividing the task of segmenting the complete, the core, and the enhancing tumor. But the limitations of this model can be that because of poor segmentation of the wider area of the tumor the subsequent networks will have a limited ground truth in their input so will lead to poor overall segmentation.

The model proposed by Nema et al. [20] has shown the best segmentation result among the papers reviewed. RescueNet has maybe been able to use Generative Adversarial Network (GAN) to its advantage and has helped in working on the issue of lack of annotated Brain MRI data. A deep neural network needs a large amount of data for efficient performance. So generating relevant features helped the model. But the main

problem with GANs is that it is very difficult to converge the generator and the discriminator network loss.

Although Deep neural network has been able to self-learn feature [20]. CNN's performance is very much dependent on the Hyper-parameter tuning or the pre and post-training processes that are done. Nonetheless there generalization can be improved wherein there are not much dependent on the architecture of the model or the choices of hyper-parameters.

VII. CONCLUSION

To begin with Most of the efficient segmentation models are based on the Deep Neural Network because they self-learn features. When it comes to Medical imaging it is a different domain where the images are more complex and high dimensional.

Furthermore it should be observed that because of the complexity and dimensionality of the Medical images we need to incorporate sparsity. The sparse images of the MRI can be representative of the original images so the complexity can be handled by the models. Although deep learning models have the ability to self-learn features but this requires a lot of data volume which is not present in this field due to Health care privacy. So our aim should be to extract features maybe using textual features like local binary pattern (lbp) or edge features like histogram of oriented gradients (HOG) so that it helps the models to learn efficiently.

Finally, It should be observed that because there is no universal architecture, researchers are always finding model which gives best results. However, in recent years CNNs have shown standard architecture can be very efficient and the pre-training and post-training process if done efficiently can improve the performance. And to take steps towards a fully automated Brian MRI segmentation method we need a deep neural network that is less sensitive to hyper-parameters or how drastic performance changes happened due to architecture changes.

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